

Econ 3385: Measuring Market Power

Problem Set 8

In this problem set, you will analyze jets.csv, which reports sales, characteristics, and prices for business jets sold by the 6 leading firms. The market is divided into 3 segments: small, medium, and large. The data is collapsed at the manufacturer-year-segment level, so it is “as if” each manufacturer has at most one jet model in each segment (really the prices and characteristics are averages over several similar models)

The variables in the data are:

- **year**: year.
- **segment**: S, M, or L.
- **manuf**: manufacturer.
- **purchases**: annual sales.
- **range**: aircraft range in nautical miles.
- **vol**: internal volume (for passengers/cargo).
- **speed**: maximum speed.
- **noise**: a measure of how loud the engines are in decibels.
- **power**: engine power in kN.
- **length**: aircraft length.
- **wingspan**: wingspan.
- **maxw**: maximum loaded weight.
- **market_size**: number of potential buyers.
- **price**: price in millions of 2010 dollars.

Questions

- Your assignment is to analyze the data and write a report that answers the questions below. You should start by summarizing the main variables, presenting a model, developing an estimation strategy, then discussing the estimated model before using the model to answer the prompts below.
 - HINT: you will need to understand the “BLP instruments” from the last lecture for this to work. These can be used as instruments for price. Think about why.
 - Note that you don’t have to include all the characteristics in your demand model. What do you need to include? What seems to matter?
 - This will take time, so don’t leave it to the last minute!
1. Measure the profit margins of each firm, each year, and show how they change over time. Decompose changes in profits into changes in costs and changes in prices.
 2. How much would marginal costs have to decline for a merger of Cessna and Bombardier to lower prices in the year 2000?

3. Suppose Bombardier is going to introduce a new product in the year 2000. If development costs are the same, in which segment should it enter the new product? Assume that the characteristics of the new product are equal to the average characteristics of the existing products in the segment. If Bombardier already has a product in that segment, then keep that product and treat the new product as a second product in the same segment.
4. Repeat question 3 for Gulfstream. What do these findings suggest about firms' incentives to introduce new products?